

Basal Area Growth of Sugar Maple in Relation to Acid Deposition, Stand Health, and Soil Nutrients

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ABSTRACT

Previous studies have shown in noncalcareous soils that acid deposition may have increased soil leaching of basic cations above the input rate from soil weathering and atmospheric depositions. This phenomenon may have increased soil acidity levels, and, as a consequence, may have reduced the availability of these essential nutrients for forest growth. Fourteen plots of the Forest Ecosystem Research and Monitoring Network in Québec were used to examine the relation between post-industrial growth trends of sugar maple (*Acer saccharum* Marsh.) and acid deposition (N and S), stand decline rate, and soil exchangeable nutrient concentrations. Atmospheric N and S deposition and soil exchangeable acidity were positively associated with stand decline rate, and negatively with the average tree basal area increment trend. The growth rate reduction reached on average 17% in declining stands compared with healthy ones. The results showed a significant sugar maple growth rate reduction since 1960 on acid soils. The appearance of the forest decline phenomenon in Québec can be attributed, at least partially, to soil acidification and acid deposition levels.

SYMPTOMS OF SUGAR MAPLE decline have been observed at different scales in the deciduous forest of the province of Québec since the late 1970s (Daoust et al., 1993). Gradual loss of vigor involving reduced growth rate and increased susceptibility to secondary biotic and abiotic stresses typically ensue (Manion and Lachance, 1992). However, there is still considerable disagreement on the nature and causes of this decline. Many hypotheses have been advanced to explain the phenomenon, including climatic factors (Holdaway, 1990; Houle, 1990; Payette et al., 1996), forest management (Côté and Ouimet, 1996), secondary insects and fungi (Bauce and Allen, 1991; Gross, 1991), and stand dynamics (Bauce and Allen, 1991). Poor soil quality due to low soil nutrient availability and aluminum toxicity were also deemed as causal factors of sugar maple decline in Québec (Roy et al., 1985; Bernier and Brazeau, 1988a,b; Bernier et al., 1989; Hendershot and Jones, 1989; Ouimet and Camiré, 1995; Moore et al., 2000). Some scientists have linked the phenomenon partly to acidic atmospheric deposition, although there has been little direct evidence of cause and effect relationships so far (McLaughling et al., 1985; Vogelmann et al., 1985).

Many studies have shown that, for noncalcareous soils, acid deposition has increased leaching losses of basic cations above the replenishment rate from soil mineral weathering and atmospheric depositions, causing a reduction in the availability of mineral nutrients

(Ulrich, 1986; Federer et al., 1989; Foster et al., 1989; Eckstein and Hau, 1992; MacDonald et al., 1992; Mitchell et al., 1992; Morrison et al., 1992; Robarge and Johnson, 1992; Kirchner et al., 1992; Miller et al., 1993; Likens et al., 1994, 1996, 1998; Hallett, 1996; Houle et al., 1997; Markewitz et al., 1998; Friedland and Miller, 1999; Lawrence and Huntington, 1999; Ouimet et al., 2001).

A necessary element in the characterization of the tree decline phenomenon is the observation of reduced growth (Hyink and Zedeker, 1987; Ouimet and Fortin, 1992). However, this trend can be difficult to detect. Because tree and stand maturation cause long-term progressive decline in trees annual ring widths at breast height, the use of unstandardized tree ring width is problematic in forest decline studies (LeBlanc, 1990). Conversion of ring increment data to basal area increment (BAI) has been suggested as a means of quantifying growth (Phipps, 1984; LeBlanc 1990). Phipps and Whifton (1988) and LeBlanc (1990) showed that age-related trends in unstandardized BAI are generally positive, culminating at an asymptotic level that can be maintained for many decades for white oak (*Quercus alba* L.) and red spruce (*Picea rubens* Sarg.). A negative trend in BAI is thus a strong indicator of a true tree growth decline (LeBlanc, 1990).

The purpose of this study was to use BAI to examine the post-industrial growth of sugar maple trees in natural stands of Québec in relation to acidic atmospheric deposition, forest stand health, and soil exchangeable nutrient pool. We wanted to test the hypothesis that the observed growth trend of trees since the 1960s was related to soil nutrient concentrations or atmospheric acid deposition.

MATERIAL AND METHODS

Study Sites

The study area included 14 natural deciduous stands of the Forest Ecosystem Research and Monitoring Network (Réseau d'étude et de surveillance des écosystèmes forestiers [RESEF]) in Québec (Fig. 1). The study area extends from approximately 45.2 to 48.2°N and 68.0 to 79.1°W. The sites (0.50 ha) represent the regional natural vegetation types for the unevenaged northern hardwood forest, which consists mainly of sugar maple, yellow birch (*Betula alleghaniensis* Britt.), and American beech (*Fagus grandifolia* Ehrh.). Elevation ranges from 105 to 530 m above mean sea level, slope from 0 to 20%, and annual precipitation from 840 to 1130 mm. Soils are all classified as Haplorthods or Placorthods (Soil Survey Staff, 1998), or podzols or brunisol (Site 702) (Canada Soil Survey Committee, 1992). Selected site characteristics of the plots are

Abbreviations: BAI, basal area increment; BS, base saturation; DBH, tree diameter at breast height; RESEF, Forest Ecosystem Research and Monitoring Network (Réseau d'étude et de surveillance des écosystèmes forestiers).

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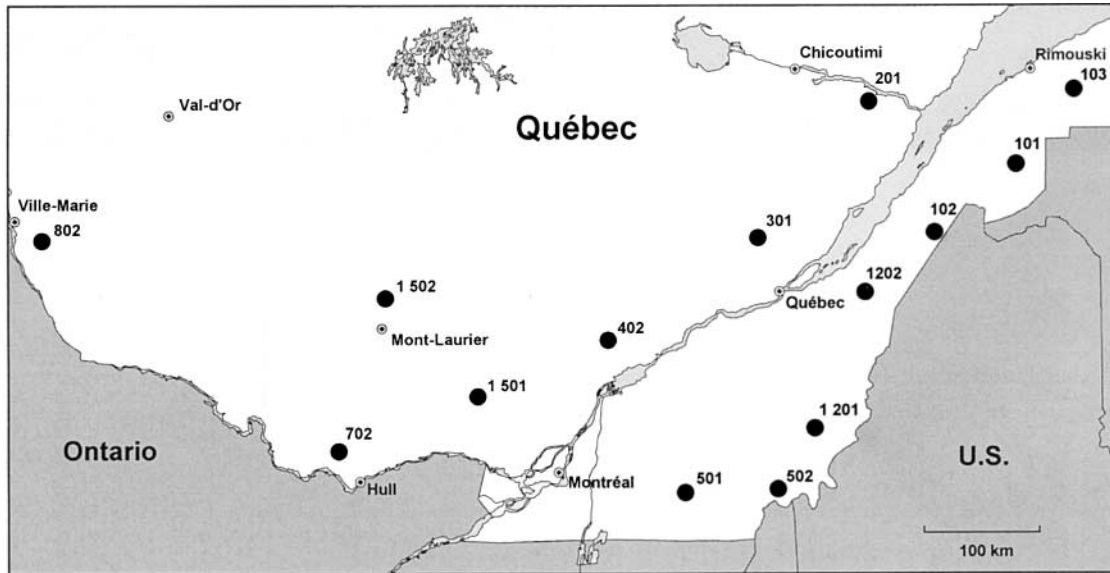


Fig. 1. Location of the Forest Ecosystem Research and Monitoring Network (Réseau d'étude et de surveillance des écosystèmes forestiers [RESEF]) study plots.

shown in Table 1. Meteorological conditions and weekly wet S and N deposition have been recorded at locations within 5 km from each plot (Boulet and Jacques, 1992, 1993a,b, 1995). Atmospheric depositions and stand characteristics are averages for the 1989 to 1993 period.

Tree Health Status

Forest health was monitored annually in the RESEF plots during the 1987 to 1995 period. Every dominant or codominant tree located in the study plot was classified by damage class (at a 5% scale) considering the dead portions of the tree canopy (Anonymous, 1989). Tree health decline classification was made by three trained foresters. Health decline classes were averaged by stand for the nine-year period.

Soil Sampling

Soils of the RESEF plots have been sampled periodically on a five-year time interval since 1987. At each sampling, six 1-m² soil pits were dug, at the plot boundaries, to a depth of 1 m or to bedrock. A soil profile description was made for each pit according to the standard Canadian method (Day, 1983). The following characteristics were recorded for each

soil horizon (St-Laurent et al., 1995): horizon depth, texture class, size and coarse fragment distribution, soil color, soil structure, and root abundance. All soil horizons with thickness greater than 4 cm were sampled individually for laboratory analysis. All soil samples were air-dried and sieved to 2 mm. Total N content was determined by automatic colorimetry (Kalra and Maynard, 1992). Exchangeable cations (K, Ca, Mg, and Al) were extracted with an unbuffered NH₄Cl (1 M, 12 h) solution, and measured with inductively coupled plasma emission. Exchangeable acidity was determined by summing the H⁺ and Al concentrations of the extract. Average concentrations of surface organic horizon and the first B horizon were used in this study.

Tree Core Sampling and Measurement

Thirty-five to 50 intact, dominant and codominant (crown forming or extending above the general level of canopy), sugar maple trees were selected at the border of each plot. Two increment cores (6-mm diameter) were extracted at breast height (1.3 m above ground level) from the opposite sides of each tree during the 1996 to 1998 period. Cores were dried and mounted in wooden blocks for sanding. Ring widths of all cores were measured to the nearest 0.01 mm under 40×

Table 1. Selected characteristics of the deciduous Forest Ecosystem Research and Monitoring Network (RESEF) sites.

Site	Elevation	Slope	Soil parent material	Soil rooting zone	Precipitation	Deposition		Average decline class	Average DBH†	Stand density†	Basal area†
						N	S				
	m	%		cm	mm	– mol. ha ⁻¹ yr ⁻¹ –		%	cm	stems ha ⁻¹	m ² ha ⁻¹
101	310	5	shale	35	900	260	280	3.6	23.2	619	30.3
102	490	0	shale	30	870	360	330	6.8	28.5	375	33.3
103	335	10	sedimentary	40	1030	250	250	2.0	21.7	651	30.3
201	105	5	acid granite	55	880	680	590	4.2	22.6	386	20.3
301	285	10	gneiss	65	1060	540	510	18.1	27.4	380	26.8
402	275	10	gneiss	40	840	540	530	2.3	24.8	445	25.4
501	350	10	ophiolitic	35	1120	360	360	5.7	26.0	520	32.0
502	480	0	schist	45	1130	410	380	7.4	25.6	415	25.9
702	215	15	acid granite	60	890	500	470	4.7	25.7	433	26.3
802	370	10	gneiss	40	870	230	280	12.3	22.8	573	27.7
1201	530	10	schist	40	930	250	250	8.6	23.5	497	25.7
1202	420	5	schist	50	990	250	290	13.2	25.6	429	28.6
1501	415	15	gabbro	55	900	280	320	7.2	26.4	403	27.5
1502	310	20	schist	45	900	410	380	4.2	23.8	561	30.2

† DBH, tree diameter at breast height. For merchantable trees with DBH ≥ 9.1 cm.

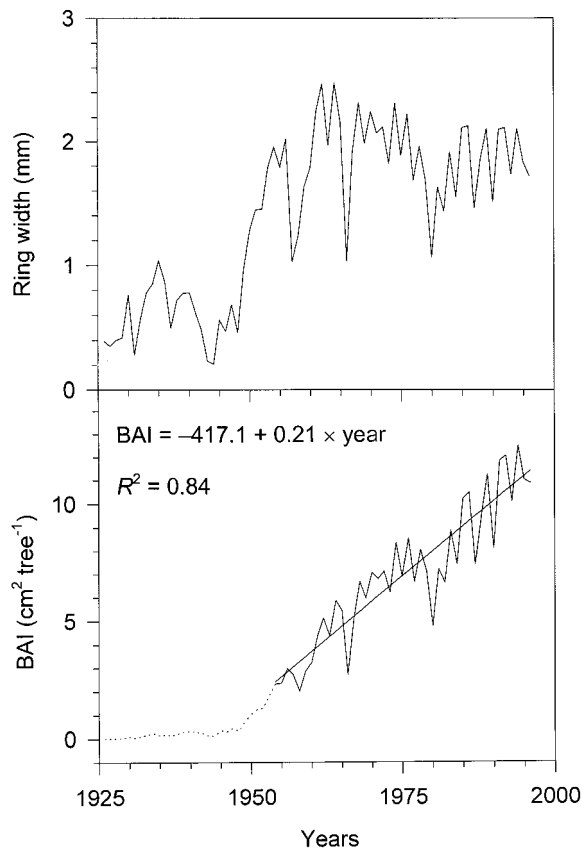


Fig. 2. Typical example (Plot 102) of sugar maple tree-ring increment and basal area increment (BAI) chronology. The dotted line corresponds to the suppression period, leaving linear trend of BAI associated to tree release.

magnification with a Velmex (Bloomfield, NY) micrometer coupled with an Acurite III digital meter, and cross-dated visually by visual examination of tree-ring sequence before measurement (skeleton plot procedure) (Fritts, 1976; Yamaguchi, 1991). Missing and absent rings were added when necessary according to standard procedure. Data were subsequently verified statistically with the COFECHA program (Holmes, 1996). Cores were rejected when they could not be cross-dated accurately. A total of 349 trees (698 cores, 14 to 47 by site) were used for the subsequent analyses. The variation in tree sample size was not due to stand structure, but to staff availability to carry on tree ring sampling and measurements.

Tree Growth Assessment

Conversion of ring width to BAI helps to remove variation in radial growth attributable to increasing circumference (LeBlanc et al., 1992). For comparing growth patterns, ring increments were thus converted to BAI with the formula:

$$BAI = \pi(R_n^2 - R_{n-1}^2)$$

where R is the tree radius and n is the year of tree ring formation.

Sugar maple is a shade-tolerant hardwood and is consequently subjected to suppression while staying in the understory. We limited the analysis to those years for which suppression was minimal. Suppression periods are characterized by very small BAI values, a slope of BAI trend near zero, and little year-to-year variation (LeBlanc et al., 1992). The suppression period was removed from the chronologies by graphical analysis. The following criterion, which was be-

lieved to be closely correlated to the year of tree release from suppression, was retained: the first year of the first 20-yr period with $BAI \geq 2 \text{ cm}^2$ for every year (Fig. 2).

Typical examples of sugar maple tree-ring and BAI chronology are presented in Fig. 2. From this figure, it appears very difficult to distinguish a clear growth pattern with ring-width measurement. However, when plotted as BAI, the pattern of tree growth becomes more apparent. Starting at the tree release, that is, at the stage that trees are freed from suppression, the BAI curve approximates a straight line according to linear regression analysis. Thus, BAI appears a better mean than ring width for quantifying tree vigor and health, as suggested by Phipps (1984), Phipps and Whiton (1988), and LeBlanc (1990).

The growth data were not standardized to preserve the growth trend over the study period. Standardization is the process of detrending ring-width data and scaling year-to-year variation to produce a variance that is more homogenous over time (Fritts, 1976). This is usually done by smoothing the data by an appropriate mathematical model, such as cubic spline, linear regression, or polynomial regression, and then calculating indices by dividing each ring width by the corresponding value of the model. Given that soils and atmospheric depositions are factors that may have long-term cumulative effects on tree growth, the unstandardized BAI contains the sought information.

Statistical Analysis

Before 1960, we assumed that regional acidic atmospheric deposition in eastern North America was not sufficient to affect tree growth (Turk, 1983). For each stand, an average unstandardized ring basal area chronology was generated by averaging BAI of each tree over each year. To prevent chronologies from shifts attributable to variation in sample size, series were furthermore truncated to keep only years where all tree samples had data. A linear regression was performed on the chronology for the 1960 to 1996 and 1998 periods or for the entire chronology when the first year was after 1960. The slope of the regression corresponded with the average tree BAI trend since 1960. Linear correlation analyses were performed between average tree BAI trend, annual atmospheric N and S wet deposition, stand health decline class, and soil nutrient concentrations. To infer average BAI trend to stand health status, tree samples must reflect stand conditions. A partial validation of this premise was attempted by comparing tree diameter at breast height (DBH) and height between stand and sample trees by t test means comparisons.

RESULTS

Soil Nutrient Concentrations

Table 2 shows nutrient concentrations in the forest floor and first mineral B layers for the study plots. In the forest floor, total N ranged from 9.3 to 19.8 g kg^{-1} , exchangeable basic cations from 6.7 to 34.5 $\text{cmol}_c \text{ kg}^{-1}$, exchangeable acidity from 0.14 to 7.7 $\text{cmol}_c \text{ kg}^{-1}$, and base saturation (BS) from 46 to 100%. In mineral B horizons, nutrient concentrations were on average 7 and 17 times lower than the humus nutrient concentrations for total N and exchangeable basic cations, respectively. Total N concentrations ranged from 1.3 to 2.9 g kg^{-1} , exchangeable basic cations from 0.5 to 3.2 $\text{cmol}_c \text{ kg}^{-1}$, exchangeable acidity from 0.9 to 9 $\text{cmol}_c \text{ kg}^{-1}$, and BS from 5.7 to 69.9%. These values are within the range observed for sugar maple stands of Québec (Ouimet and Camiré, 1995).

Table 2. Average total nitrogen and exchangeable mineral nutrients in the organic and first mineral B layers in the Forest Ecosystem Research and Monitoring Network (RESEF) plots. The standard error is given in parentheses.

Site	Forest floor						First mineral B layer					
	N	H + Al [†]	K	Ca	Mg	BS [‡]	N	H + Al	K	Ca	Mg	BS
	g kg ⁻¹	cmol _c kg ⁻¹				%	g kg ⁻¹	cmol _c kg ⁻¹				%
101	19.8 (0.84)	0.1 (0.05)	1.8 (0.13)	28.9 (2.1)	3.9 (0.2)	99 (8)	1.8 (0.09)	3.6 (0.4)	0.06 (0.01)	0.7 (0.1)	0.10 (0.01)	20 (3.6)
102	17.9 (0.71)	0.7 (0.13)	1.3 (0.11)	19.2 (1.8)	2.5 (0.1)	97 (11)	2.1 (0.16)	3.4 (0.6)	0.08 (0.01)	2.2 (0.7)	0.19 (0.04)	42 (14.0)
103	16.4 (0.87)	0.2 (0.04)	1.7 (0.10)	23.0 (2.2)	3.7 (0.2)	99 (11)	1.9 (0.08)	4.1 (0.2)	0.11 (0.01)	1.7 (0.2)	0.38 (0.05)	34 (3.8)
201	11.5 (1.70)	1.8 (0.80)	1.4 (0.19)	15.4 (2.9)	3.1 (0.7)	92 (19)	1.6 (0.36)	1.7 (0.8)	0.07 (0.02)	2.0 (0.5)	0.38 (0.13)	59 (18.9)
301	10.5 (2.62)	7.7 (0.96)	0.8 (0.09)	5.1 (1.1)	0.8 (0.1)	46 (9)	2.1 (0.54)	5.0 (0.5)	0.11 (0.02)	0.5 (0.1)	0.11 (0.02)	12 (1.7)
402	16.0 (0.63)	4.2 (0.92)	1.0 (0.05)	15.5 (1.3)	2.7 (0.2)	82 (8)	2.9 (0.24)	3.3 (0.4)	0.07 (0.01)	0.7 (0.1)	0.11 (0.02)	20 (4.0)
501	18.0 (0.69)	1.3 (0.36)	0.9 (0.08)	18.3 (2.1)	11.4 (1.7)	96 (12)	1.3 (0.23)	1.4 (0.4)	0.05 (0.02)	1.5 (0.4)	1.61 (0.19)	70 (13.2)
502	17.9 (0.83)	3.9 (0.48)	1.4 (0.13)	8.2 (0.5)	1.6 (0.1)	74 (5)	2.8 (0.23)	6.0 (0.3)	0.10 (0.01)	0.4 (0.1)	0.11 (0.01)	8 (0.9)
702	9.7 (0.79)	0.5 (0.21)	0.7 (0.10)	18.1 (2.7)	2.6 (0.3)	98 (17)	1.5 (0.17)	0.9 (0.1)	0.05 (0.01)	1.2 (0.2)	0.17 (0.04)	62 (11.5)
802	9.3 (0.57)	0.9 (0.15)	2.0 (0.24)	8.4 (0.6)	1.6 (0.6)	93 (7)	2.6 (0.23)	3.4 (0.3)	0.07 (0.01)	0.7 (0.1)	0.10 (0.01)	19 (3.7)
1201	19.6 (0.24)	1.2 (0.27)	1.5 (0.06)	17.0 (0.9)	4.0 (0.3)	95 (6)	2.4 (0.07)	5.3 (0.5)	0.10 (0.01)	1.2 (0.2)	0.34 (0.07)	23 (3.7)
1202	17.3 (1.19)	2.4 (0.30)	1.6 (0.14)	7.72 (0.7)	1.8 (0.1)	82 (7)	1.9 (0.19)	9.0 (1.0)	0.08 (0.01)	0.4 (0.1)	0.10 (0.01)	6 (1.0)
1501	9.7 (1.00)	2.9 (0.45)	1.1 (0.39)	10.0 (2.6)	1.3 (0.2)	81 (22)	2.5 (0.13)	3.5 (0.2)	0.07 (0.01)	0.7 (0.9)	0.10 (0.02)	21 (2.2)
1502	9.5 (0.51)	2.5 (0.49)	0.9 (0.25)	8.8 (0.5)	1.4 (0.1)	82 (6)	2.0 (0.16)	2.8 (0.2)	0.06 (0.01)	0.5 (0.4)	0.80 (0.01)	19 (1.8)

[†] Exchangeable acidity.[‡] Base saturation.

Growth Trend

Mean age of sampled trees in the RESEF plots ranged from 71 to 106 yr, while age since release ranged from 51 to 75 yr (Table 3). On average, trees were released from the suppression phase 25 yr after they grew to a height of 1.3 m above the ground. Average unstandardized BAI chronologies of sugar maple and the associated linear BAI trend since 1960, after removal of the suppression period, are presented in Fig. 3. Average tree BAI trends since 1960 ranged from -0.33 to 0.11 cm² yr⁻¹. The analysis indicated that, on average, trees in 6 of the 14 study sites showed a negative growth trend, suggesting a continuous growth decline since 1960. For stands showing a negative trend, the growth rate change at the onset of decline appeared to be relatively abrupt, generally beginning in a narrow time window from 1950 to 1965. Within each stand, however, the variability in BAI trend among trees was high, as expressed by the minimum and maximum values (Table 3).

Correlation Analyses

Mean DBH and height comparisons between the whole tree population in the stand and the cored domi-

nant and codominant tree population are presented in Table 4. With respect to these observations, most of the sampled tree populations were representative of stand populations, except for one site where mean sample DBH was significantly smaller than average stand DBH, and two sites where mean sample height was significantly different from mean stand height. Overall, these results indicate that the sampled tree populations were not biased compared with stand tree populations.

Correlation analyses between BAI trend, stand health status, atmospheric deposition, and soil element concentrations are presented in Table 5. The BAI trend was negatively related to atmospheric N and S wet depositions, stand health decline class, forest floor exchangeable acidity, and exchangeable K concentration in the mineral B horizon. The BAI trend was, however, positively related to the forest floor BS. Stand health decline classes were positively related to exchangeable acidity in both soil layers, and negatively associated with exchangeable Ca and BS in the forest floor. Current rates of atmospheric N and S deposition are positively associated with levels of soil exchangeable acidity, and negatively with exchangeable K concentration in the forest

Table 3. Mean age and unstandardized basal area increment (BAI) trend of sugar maple chronologies since 1960 and their associated standard error (SE).

Site	Number of trees sampled	Tree age at DBH [†]		Age since release		Linear BAI trend of average unstandardized chronology since 1960		Range in linear BAI trend of individual tree chronologies since 1960	
		Mean	SE	Mean	SE	Slope	SE	Minimum	Maximum
		yr				cm ² tree ⁻¹ yr ⁻¹			
101	14	73	4	51	2	0.06	0.02	-0.02	0.57
102	15	79	5	56	3	0.11	0.02	0.04	0.40
103	47	75	4	51	2	0.01	0.05	-0.08	0.35
201	17	104	8	63	5	-0.08	0.02	-0.07	0.24
301	39	103	4	75	4	-0.33	0.02	-0.57	0.25
402	32	87	5	54	4	-0.06	0.02	-0.12	0.37
501	25	86	4	68	3	0.002	0.02	-0.23	0.23
502	28	73	3	60	3	-0.16	0.02	-0.43	0.22
702	15	106	6	69	4	0.03	0.03	-0.15	0.52
802	15	97	6	75	6	0.05	0.03	-0.08	0.43
1201	21	82	5	58	4	-0.06	0.05	-0.12	0.36
1202	30	78	5	53	4	-0.02	0.02	-0.30	0.42
1501	16	95	5	66	4	0.03	0.03	-0.07	0.31
1502	35	71	4	57	3	0.04	0.02	-0.02	0.39

[†] DBH, tree diameter at breast height.

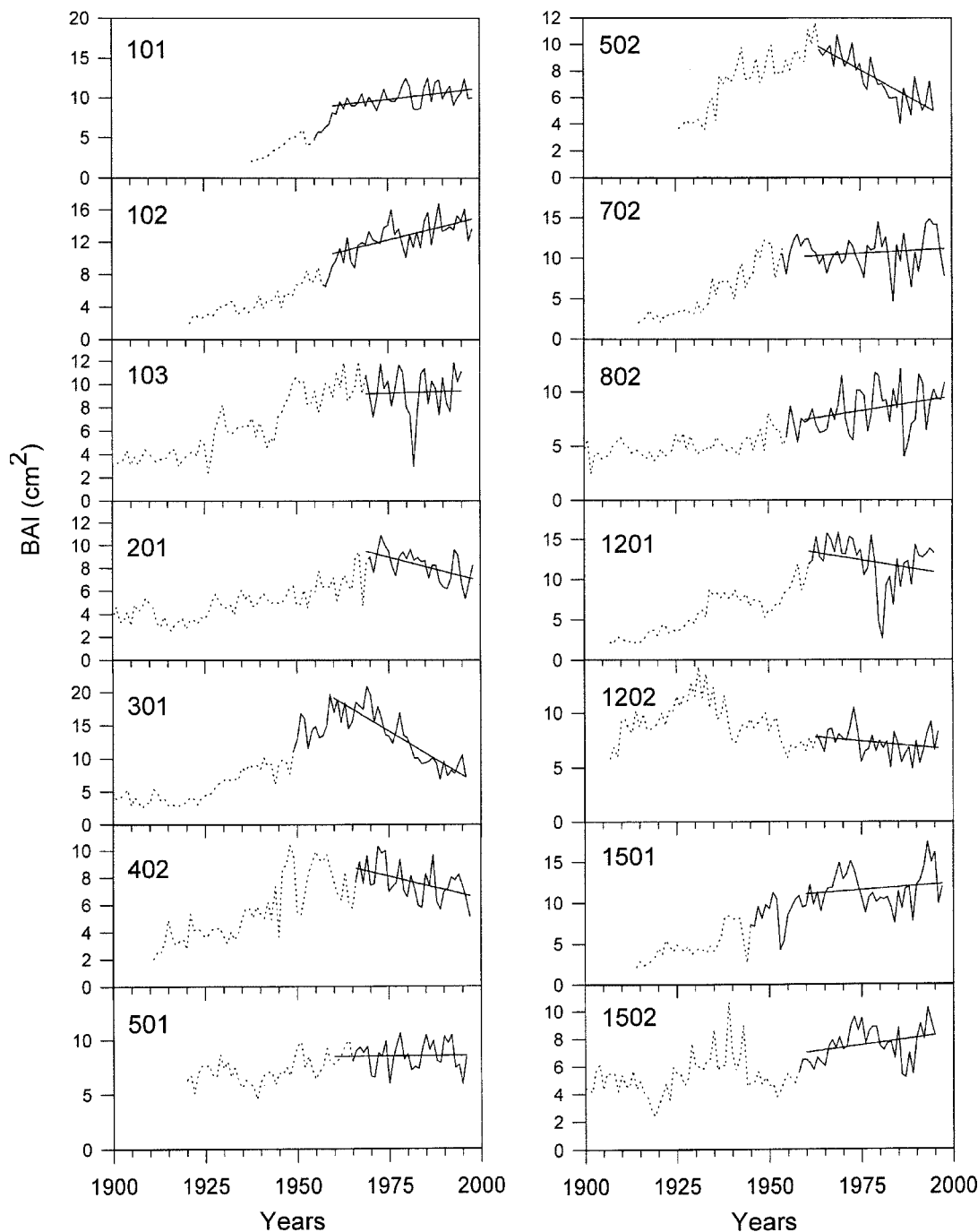


Fig. 3. Average unstandardized basal area chronology of sugar maple for deciduous Forest Ecosystem Research and Monitoring Network (Réseau d'étude et de surveillance des écosystèmes forestiers [RESEF]) study plots in Québec after removal of the suppression period, and associated linear growth trend since 1960. Early parts of series (dotted line) represent the reduced number of cores.

floor. Most significant relationships between average BAI trend of dominant and codominant trees since 1960 and forest floor characteristics are shown in Fig. 4.

DISCUSSION

Natural BAI decline may be associated to senescence or individual competition. Sugar maple trees have a life expectancy of more than 300 yr (Godman, 1965) while mean age of the sample population ranged from 71 to

106 yr. Moreover, the average BAI trend is not correlated to the average tree age ($p = 0.29$) or even the average tree age since release ($p = 0.23$), indicating that the observed growth trend was not attributable to age maturation. The observed BAI trend also was not correlated to the number of merchantable stems by hectare ($p = 0.14$). Despite the fact that individual competition indexes were not available, we are confident that the observed growth trend at the stand level was not caused by inter-individual competition. This is in accordance

Table 4. Comparison of means (*t* test) between sample and stand trees for DBH† and height.

Site	Stand DBH		Sample DBH		DBH <i>p</i> for <i>t</i> test	Stand height		Sample height		Height <i>p</i> for <i>t</i> test
	Mean	CI‡	Mean	CI		Mean	CI	Mean	CI	
101	26.3	12.1	27.0	3.5	0.77	20.3	0.2	21.3	1.1	0.04
102	38.4	3.0	29.9	3.2	0.07	20.4	0.4	20.1	0.9	0.62
103	26.6	1.6	26.7	2.9	0.94	19.1	0.2	20.1	0.5	0.002
201	32.5	2.4	27.5	4.1	0.06	19.8	0.5	19.0	0.9	0.16
301	33.0	1.8	35.1	4.1	0.33	21.9	0.4	22.5	0.9	0.17
402	30.0	1.9	26.7	2.9	0.11	21.3	0.4	18.8	0.7	<0.001
501	29.5	1.5	29.0	3.9	0.81	23.5	0.4	22.9	1.2	0.34
502	29.0	1.8	26.1	3.9	0.16	20.8	0.5	20.2	1.1	0.24
702	27.8	1.5	32.6	1.5	0.09	23.1	0.4	23.2	1.3	0.91
802	26.7	1.1	29.3	4.9	0.20	20.0	0.2	20.7	1.4	0.07
1201	30.7	1.4	31.3	4.3	0.76	20.2	0.2	20.7	1.1	0.08
1202	32.7	2.6	24.3	4.7	0.002	18.2	0.3	17.3	1.3	0.06
1501	32.1	2.0	33.0	4.4	0.77	21.5	0.4	20.6	1.0	0.12
1502	28.7	1.4	25.6	2.9	0.06	20.7	0.3	21.0	0.8	0.41

† DBH, tree diameter at breast height.

‡ CI, 95% confidence interval.

with findings of Ouimet and Fortin (1992), who did not observe relationships between the competition index and tree growth rate in declining stands.

Examination of the maximum and minimum BAI trend of individual trees indicates that the tree growth trend was variable within a given site (Table 3). This finding suggests that stand or soil conditions exceeded the tolerance limit of some trees, so they may have modified their radial growth rate in the long term. The variability in tree BAI trend within site may be attributable to spatial variability of stand or soil conditions. The tolerance limit of each tree may be also attributable to individual genotype.

A negative trend in BAI is a strong indicator of a true growth decline (LeBlanc, 1990). Based on this criteria, the BAI trend of trees located in declining stands was about -13.8% per year while, on average, the BAI trend of trees located in healthy stands increased annually by 3.5% . These results suggest that atmospheric acid depositions and the resulting soil acidification or mineral nutrient depletion are related to tree BAI growth rates, the latter decreasing by about 17% ($-13.8 - 3.5 = -17\%$) since the 1960s in comparison with growth rate of trees growing in healthy stands.

The fact that acid deposition is a major stress contributing to forest decline has been vigorously debated. One hypothesis suggests that acid deposition increases leaching losses of basic cations, resulting in imbalances in nutrient availability on sensitive sites (Ouimet and Camiré, 1995). Another suggests that soil acidification induced by acid rain could increase soil acidity to levels that may interfere with root growth and nutrient uptake and tree growth (e.g., Ouimet et al., 1996). The significant relationship between stand health decline classes and soil exchangeable Ca ($r = -0.68, p = 0.008$) suggests that the pool size of basic cation reached deficiency levels for some forest stands. Moreover, the relations between BAI trend, health decline class, and soil exchangeable acidity concentrations support the hypothesis that soil acidity may interfere with tree growth and health. This hypothesis is corroborated by positive maple growth and health response to liming (Moore et al., 2000; Long et al., 1997). In these ecosystems, acidic atmospheric deposition and soil acidification may have caused depletion of exchangeable basic cations in soils

with limited mineral weathering rates (Ouimet et al., 2001). This situation may result in severe effects on sugar maple productivity in the very near future.

In accordance with Phipps and Whiton (1988), the results of this study suggest that the BAI trend is related to soil and site quality. In accepting this hypothesis, it would be expected that the BAI trend would continue in a linear fashion as long as the stand is not disturbed. If a tree is subjected to a punctual stress (climatic extreme, partial defoliation, crown released by partial cutting, etc.), tree growth rate can be rapidly modified for a short time and then resume growth at nearly the same rate as before. Thus, the BAI trend would be expected to be the same, both before and after punctual stresses; only during negative effects of punctual stress would the BAI trend be temporarily modified. A prolonged growth rate change would imply either a change in site quality (including atmospheric deposition) or a long-

Table 5. Linear correlation coefficients between basal area increment (BAI) trend of the average unstandardized chronology, decline class, N and S deposition, and soil nutrient pool of the forest floor and first mineral B horizon ($n = 14$).

	Average BAI trend	Stand decline class	Atmospheric deposition	
			N	S
Stand decline class	-0.54*			
Atmospheric deposition				
Total N	-0.48†	-0.10		
Total S	-0.49†	-0.05		
Forest floor				
Total N	0.09	-0.21	-0.36	-0.42
H + Al‡	-0.85***	0.56*	0.46†	0.52†
K	0.28	0.06	-0.59*	-0.61*
Ca	0.50	-0.68**	-0.17	-0.23
Mg	0.20	-0.31	-0.10	-0.12
BS§	0.84***	-0.66**	-0.35	-0.41
First mineral B horizon				
Total N	-0.20	0.15	-0.14	-0.09
H + Al	-0.32	0.54*	-0.41	-0.38
K	-0.63*	0.42	-0.11	-0.15
Ca	0.34	-0.42	0.20	0.09
Mg	0.07	-0.18	-0.01	-0.03
BS	0.30	-0.45	0.36	0.29

* Significant at the 0.05 probability level.

** Significant at the 0.01 probability level.

*** Significant at the 0.001 probability level.

† Significant at the 0.10 probability level.

‡ Exchangeable acidity.

§ Base saturation.

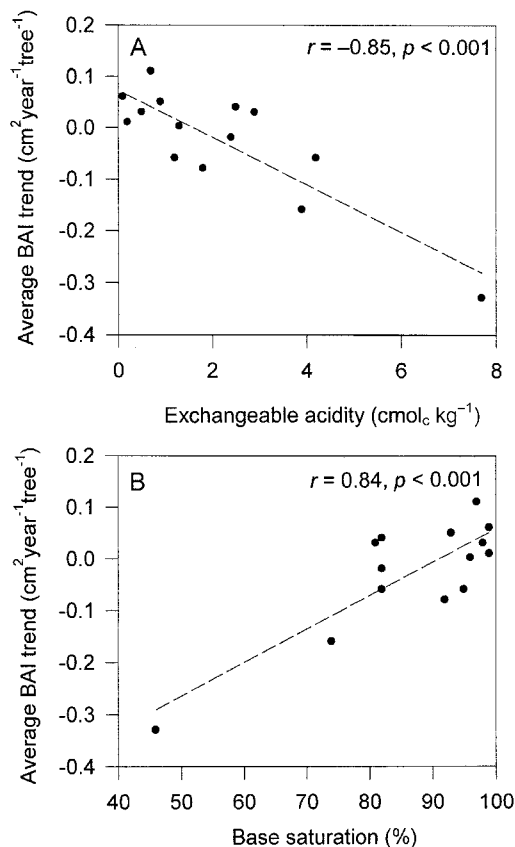


Fig. 4. Average basal area increment (BAI) trend of dominant and codominant trees since 1960 as a function of (a) exchangeable acidity and (b) base saturation of the forest floor in 14 sugar maple stands.

lasting change in tree growth physiology and metabolism.

Smith (1974) and Bondietti and McLaughlin (1992) suggested that, in advanced stages of forest response to air pollutant, plants would become increasingly sensitive to other biotic and abiotic processes. Previous studies have related sugar maple decline to a succession of punctual factors like defoliators or climatic anomalies (e.g., Payette et al., 1996; Bauce and Allen, 1991). These results relating initiation of decline to historical punctual stress are consistent with the advanced stage of acid precipitation effects where plants become increasingly sensitive to other biotic and abiotic stress factors (Houston, 1999; Horsley et al., 2000).

CONCLUSION

This study attempted to relate post-industrial growth trends of sugar maple to acid deposition, soil fertility, and forest health in natural stands. Higher atmospheric N and S deposition and soil acidity were related to stand health decline rate and a decreased average of tree BAI trend. The BAI trend reduction of trees reached on average -17% in declining stands compared with healthy ones. Although some relationships exist between these variables, cause-effect relationships cannot be tested in such an exploratory study. Given the impalpability of the acid deposition effects on natural ecosys-

tems because of confounded effects, this study provides evidence of links between soil fertility, atmospheric acid deposition, and forest growth and health. Because acid deposition is a long-term cumulative stress, future dendrochronological studies on effects of atmospheric pollution should focus on growth trends instead of standardized data, which express punctual variability of environmental factors. Further studies are needed to expand the relation we found between BAI trends and soil fertility.

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